1. Twist points cannot supply index-calculus relations: a measured validation

Study `research/twist_obstruction_20261026/`, 2026-10-26.

- Run: `reports/run-20261026T1456Z/`. The run report is `RUN_REPORT.pdf`.
- Protocol: `PROTOCOL.md`, frozen at commit `ff3bf9d48` before the run.
- Harness: `src/main.rs`, native Rust with std only.
- Aggregation: `summarize/`, native Rust.
- Plotting: `plot.py`, layout only.

Each statement below is tagged **theorem**, **exact computation**, **heuristic reference** or **not established**.

1.1. Question

Take E and its quadratic twist E′. They share x-coordinates, so they share summation polynomials. Can the twist's points feed relations for the DLP on E? The argument says no. $\varphi(E'(F_q))$ is the $\sigma = -1$ part of the x-rational points of $E(F_{(q^2)})$, and $E(F_q)$ is the $\sigma = +1$ part. So $R = A + B$ forces $2B = 0$.

This study makes three consequences of the argument concrete and measures them in the ECC2K-130 model: $y^2 + xy = x^3 + ax^2 + b$ over $F_{(2^n)}$ with n odd. The twist is $a \mapsto a+1$, and $\varphi(x, y) = (x, y + sx)$ with $s^2 + s = 1$.

1.2. Statements and their status

1. **T1 (theorem).** Let $P \in E(F_q)$ and $Q \in \varphi(E'(F_q))$. Then $P + Q \in E(F_q)$ iff $Q \in E[2](F_q) = \{O, T\}$, where $T = (0, \sqrt{b})$.
   - **Exact computation:** exhaustive at n = 7, 11, 13 on 15 curves.
   - 356,530,036 sums with Q not 2-torsion; **0** are rational.
   - 103,884 sums with $Q \in \{O, T\}$; all are rational.

2. **T2 (theorem).** Let $R = Q_1 + \ldots + Q_m$ with every $x(Q_i) \in F_q$. Then the sub-sum of the Q_i whose σ-parity is opposite to R's lies in {O, T}. Consequences:
   - For m = 2 there is no mixed solution.
   - For m = 3 the mixed solutions are exactly (x_R, x, x) when $x_R \in V$, and $(x(R+T), x, x(P_x+T))$ when $x(R+T) \in V$.
   - Every F_q-root of $S_{(m+1)}$ is therefore a valid relation on E once the twist sub-sum is dropped. No root is lost.

   Exact computation: 4954 decomposition targets, n = 11 to 19, factor base $V = \text{span}(1, \ldots, t^{(l-1)})$.
   - 0 violations on both models.
   - On 1154 of 1154 m = 3 targets, the observed off-parity set equals the set predicted from the target and V alone.
   - The positive controls are detected:
     - C1: 18,042 planted (x_R, x, x) solutions, all found;
     - C2: 95 planted T-pair solutions, all found.

3. **T3 (theorem).** The x-solution set is the same on E_a and $E_{(a+1)}$. For binary curves, $S_3 = x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2 + x_1x_2x_3 + b$ does not involve a at all.
   - **Exact computation:** the x-sets are identical on 4954 of 4954 targets.
   - The point sets equal the explicit S_3 roots on 3800 of 3800 m = 2 targets, and the explicit $S_4 = \text{Res}(S_3, S_3)$ roots on 105 of 105 m = 3 targets.

4. **Heuristic reference, reported as a distance.** Suppose lift types were independent of the solution condition. Then a fraction $1 - p_E^m$ of solutions would contain a twist-lifting coordinate: 0.74–0.77 for m = 2 and 0.86–0.89 for m = 3. The observed fraction is:
   - m = 2: 0, with every 95 % upper bound ≤ 0.0016;
   - m = 3: 0.041–0.063 at n = 11, 13, and 0 at n = 17, 19.

   All nonzero m = 3 mass is the predicted degenerate family (x_R, x, x), hit because x_R happens to fall in V ($|V|/q$ = 3.1 % and 1.6 %). Those "mixed" solutions carry no new information: the relation is R = R.

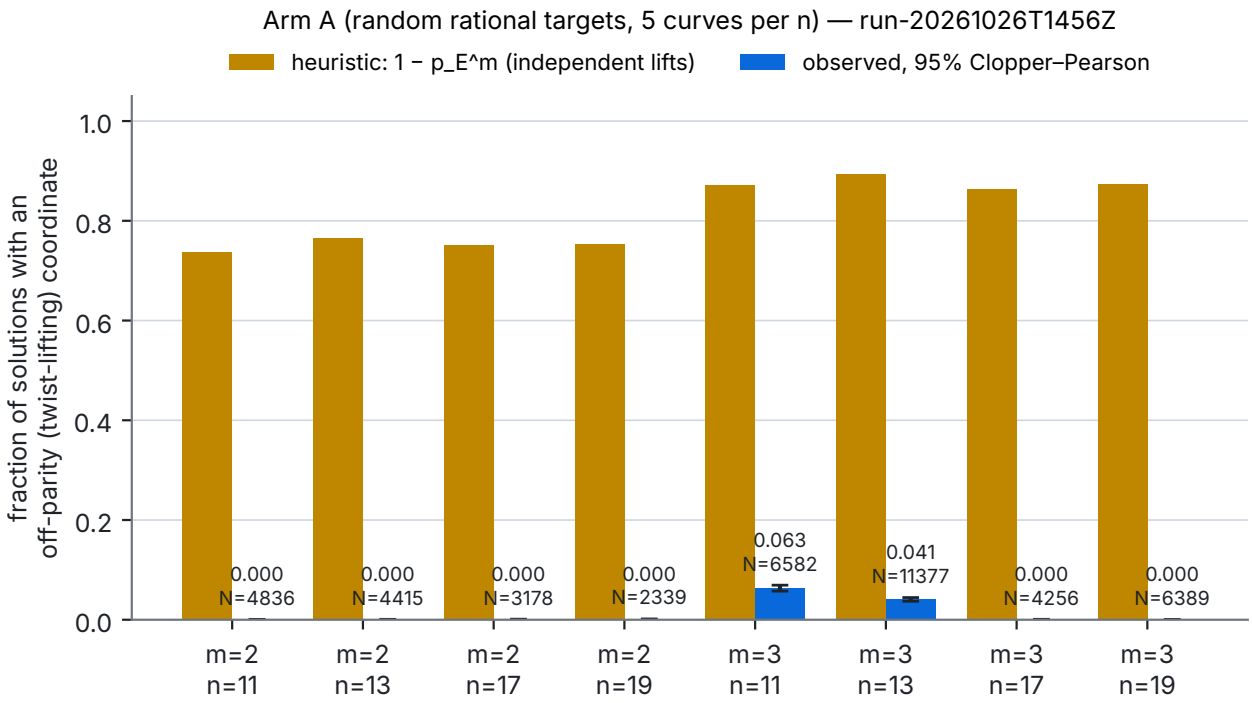


Figure 1: Arm A, run-20261026T1456Z: observed off-parity fraction (95 % CP) against the independent-lift heuristic.

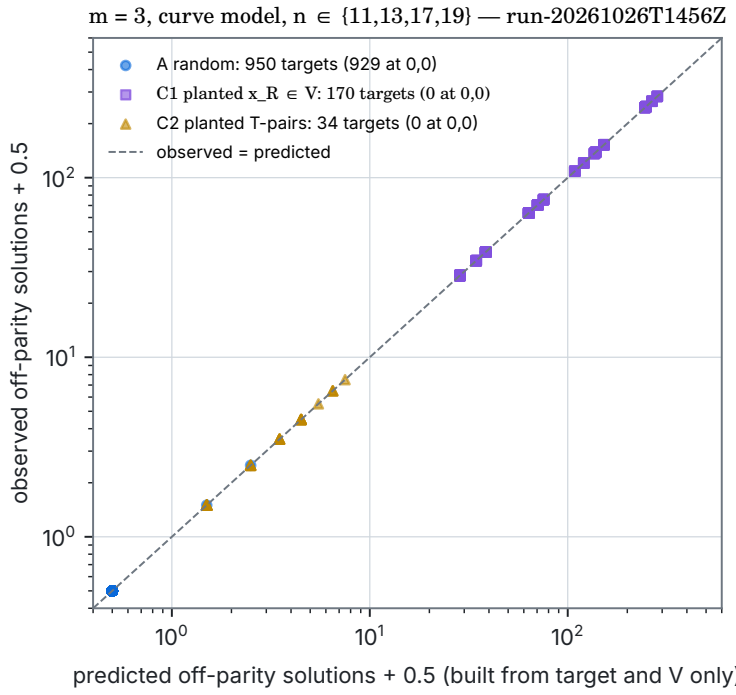


Figure 2: Predicted against observed off-parity solutions per target (m = 3). Every point lies on the diagonal.

1.3. Interpretation

The separation holds in this model. The measured behaviour matches the argument exactly, including its degenerate cases. Three practical consequences for the index-calculus programme:

- **Twist points are not a source of relations.** Adding φ(E′(F_q)) to the factor base adds columns that no non-degenerate relation on E can use. The roughly half of x ∈ V that lift to the twist contribute nothing, and the factor base loses nothing by excluding them. The same picture appeared on prime fields in `research/pair46_factor_bases_20261015/05-qr-window/`: the twist-aware filter changed the candidate counts and added no relations.
- **Rationality checking is lossless.** Every F_q-root tuple of S_(m+1) is a valid E-relation after the twist sub-sum is dropped, and for m = 2 every root tuple is pure. No y-recovery or lift-checking step in a relation pipeline discards a usable root.

- **A correctness invariant for decomposition code.** Any pipeline that reports an off-parity solution outside the predicted structured set has a bug. This study's instrument provides that check, with positive controls.

What could escape the obstruction (**not established**, no candidate known):

- Any construction built from group operations defined over F_q commutes with σ, and once 2 is inverted no nonzero equivariant map sends the −1 part to the +1 part.
- The ℓ-parts of E and E′ can coincide only when ℓ divides 2t. For $\ell > 2\sqrt{q}$ that forces t = 0, a supersingular curve, which MOV already reduces.
- The shared geometry does allow **amortisation**: one descent or Gröbner set-up serves both the DLP on E and the DLP on E′. That is an accounting change, not an improvement to a single ECDLP (CLAUDE.md success criterion).

1.4. Scope and limits

- The computation covers $n \in \{7, 11, 13\}$ (census) and $n \in \{11, 13, 17, 19\}$ (decompositions) with $m \le 3$ and the low-degree subspace V. The general statements rest on the proof. The computation shows the proof's predictions hold on every instance enumerated and that the instrument can see off-parity solutions when they are planted.
- The violation counter was exercised only by real data, never by a planted violation. It is the `else` branch of the same classifier that the C1/C2 off_O/off_T branches exercise.
- No ECDLP cost was measured, and no IC leaderboard row, benchmark row or rho comparison is made or implied. Wall times are host-dependent and recorded only for provenance.
- No `identity.certificate/v1` is issued. The identities here are group-theoretic, over F_(2^n). The v1 checker evaluates over F_(2^61−1) and cannot express them. Independence of S_3 from a is syntactic.